Blood Metals and Metabolic Syndrome: A Machine Learning and Mendelian Randomization Investigation

1. State of Problem

Ask how someone got their high blood lead levels and the answer is usually boring: home or work. Paint chips in old houses. Soil near a road that was never tested. Pipes no one got around to replacing. None of it is new either: leaded gasoline hasn't been sold for decades, but its residue still appears in exposure data. Industrial emissions add to the mélange. This risk is further heightened in workplaces built around battery recycling or metal smelting, whereby workers bring it home, often unknowingly, and expose their families in the process [1,2].

A threshold level below which lead exposure is considered safe has not been identified [3,4]. It builds up in the body and impacts several organ systems simultaneously: neurological, cardiovascular, renal, hematological. Even low levels of exposure have been linked to impaired cognitive development, behavioral problems, increased blood pressure and cardiovascular disease, while higher levels of exposure can cause kidney damage, anemia or death [3-5]. The children get the worst of it. Their nervous system is still developing. This alone puts them at greater risk and the resulting cognitive deficits can persist well into adulthood [4,6].

Lead is the best-documented of these exposures, but it is not the only one to be concerned about. Exposure to cadmium and mercury comes mainly from cigarette smoke, shellfish consumption, and industrial exposure. Both of these metals have been shown to track fairly consistently with increased odds of metabolic syndrome and several of its individual components in large observational cohorts [7-11]. Less clear is the case for selenium, which is required by the body but where blood concentrations above the physiological range have also been associated with increased prevalence of metabolic syndrome, particularly in the presence of zinc [12,13], although selenium deficiency has also been linked to metabolic syndrome [14]. Manganese is the least settled of the four, with some studies showing risk increase only in specific age groups and others none [15,16]. None of these associations behaves the same way in different populations, which is itself part of the motivation for testing all five metals side-by-side in the same cohort, rather than treating lead as a stand-in for the group.

Metabolic syndrome (MetS) is not one disease. It's a pattern: abdominal obesity, insulin resistance, high blood pressure, high triglycerides, low HDL cholesterol: all showing up together in the same patients more often than chance would predict. Each on its own increases the risk of cardiovascular disease and type 2 diabetes. But stacked together, they push that risk far beyond what you'd get by just adding the individual effects [17-21].

Most of the damage is done by insulin resistance and central obesity. Then you add on dyslipidemia and hypertension. The full mechanistic picture is not yet complete, but bad diet and lack of physical activity are clearly near the center of it [22-24].

Mortality associated with MetS, particularly cardiovascular mortality, continues to rise with its global prevalence, partly driven by parallel epidemics of obesity and physical inactivity that do not appear to be abating [20,25,26].

Blood lead is associated with MetS throughout more than one biological pathway. As lead levels increase MetS risk increases and its individual components (blood pressure, triglycerides, fasting glucose and HDL cholesterol) all go in the predicted way. A notable data point comes from the Korea National Health and Nutrition Examination Survey where doubling blood lead alone was related with a more than 1.5-fold increase in probabilities of MetS [27]. Lead also interferes with lipid metabolism and causes insulin resistance, the two pathways that are thought to be most directly responsible for MetS, and increased lead has been independently associated with increased triglycerides and fasting glucose and decreased HDL [28,29]. There is also a cardiovascular aspect: oxidative stress generated by lead seems to aggravate circulatory dysfunction and older patients with MetS seem more susceptible to autonomic issues with continuous exposure [5,30]. Thyroid function may also play a role, as thyroid-stimulating hormone has been identified as a potential mediator between lead exposure and MetS [31].

The molecular picture for the other four metals is similar: oxidative stress, systemic inflammation and altered lipid and glucose metabolism continue to emerge as the routes linking metal exposure to MetS, independent of which metal is being studied [13,32]. The difference lies in the extent to which co-exposure, genetic background, and lifestyle factors such as food and smoking bend that link in every particular group, which is likely why the amount and even the direction of the correlation vary so dramatically from one cohort to the next [33-35].

But none of this resolves the question of how strong the blood lead–metabolic syndrome link is. Several Korean studies have shown a clear association, where higher blood lead is associated with higher odds of MetS, higher triglycerides, and larger waist circumference [10,27,28]. Other studies do not see the same pattern. Findings vary across samples and age bands, and the strength of the lead association is not consistent from one study to the next, which is why the pooled estimates carry substantial heterogeneity [7,8,36]. Some data indicate that lead affects specific components of MetS (blood pressure and lipids in particular) more than it predicts the syndrome as a coherent whole [29,37].

The picture is no better for the other metals. Large cross-sectional studies and meta-analyses report inverse or null relationships for lead, cadmium and mercury in particular populations, so the direction of association is not settled even for the two metals with the most consistent overall evidence [8,16,38]. Analyses of United States populations have reported associations that differ from those in Asian cohorts in both direction and magnitude, which is a harder problem than a simple difference in effect size [33,34]. Manganese is the least clear of the group, with higher risk seen in some studies and only in certain age groups, and not seen at all in others [15,16].

Machine learning paired with SHAP (SHapley Additive exPlanations) has turned into a go-to combination for disease prediction. It is also useful for something more specific: figuring out which risk factors actually matter, and by how much. XGBoost, Random Forest, and Gradient Boosting are the usual suspects here. They have already been run against hypertension, heart disease, Alzheimer's disease, and

diabetic retinopathy: different diseases, same basic move, pulling together clinical, demographic, and lifestyle data to reach high predictive accuracy [39-42].

What ML does particularly well is rank risk factors rather than just flag them. SHAP scores each feature's contribution within a model, so researchers can see exactly which variables are doing the predictive work: cholesterol, blood pressure, and age for heart disease; hemoglobin A1c and systolic blood pressure for diabetic retinopathy [40,43,44]. The same technique visualizes how, say, age and triglycerides feed into hypertension risk, which in turn opens the door to more targeted interventions [39,45].

That interpretability matters clinically, because it is often the missing ingredient that keeps complex models out of practice. ML-SHAP combinations have already been used to predict mortality in ICU patients with pressure ulcers and to assess COVID-19 reinfection risk, offering actionable, early-warning insight in both cases [46,47].

Mendelian Randomization (MR) offers a way around the uncertainty that plagues observational studies. By using genetic variants as instrumental variables, it can test causal links between a risk factor and a health outcome without needing a randomized controlled trial, which, for something like metal exposure, simply is not feasible or ethical. Because genetic variants are assigned essentially at random during meiosis, they are distributed independently of the confounders that trouble observational designs. Using a genetic proxy for an exposure (cholesterol levels being a classic example) lets researchers check whether an observed association actually reflects a causal effect, or just correlation [48-50].

Taken together, the literature points to a plausible but unsettled role for blood metals (lead the best documented among them, alongside cadmium, mercury, selenium, and manganese) in the development of metabolic syndrome. Untangling which of these associations reflect real biology, and which reflect confounding by diet, smoking, or socioeconomic circumstance, will take more than another single-metal cross-sectional study. This proposal takes a two-step approach to that problem. First, machine learning ranks all five metals, alongside a broad panel of demographic, anthropometric, and lifestyle covariates, by how much each actually contributes to MetS risk once the others are accounted for, rather than assuming in advance which one matters most. Second, Mendelian Randomization takes whichever exposures come out of that ranking with a defensible genetic instrument (the metals, and any non-metal factor the modeling flags as comparably important, such as body mass index) and tests whether their contribution reflects a genuine causal effect or just confounding.

2. Literature Review

Age, sex, and gender all shape MetS risk. Nobody disputes that much. What is harder to pin down is exactly how they interact, since the pattern shifts by population and by study design. Prevalence rises with age in both sexes, and older adults consistently show higher rates than younger ones [26,51-53], though the shape of that rise differs by sex. Before age 50, men in many populations run a higher MetS prevalence than women; after menopause, women's rates climb sharply and often overtake men's

[54,55]. Hormonal shifts explain part of this reversal. So do gender-related factors like lifestyle and psychosocial stress [56,57].

The way MetS components cluster also splits along sex lines. Abdominal obesity tends to dominate the picture for women, whereas men's risk profiles are more mixed, often skewing toward elevated blood pressure and triglycerides at younger ages [52,54,56]. Age hits women harder in odds-ratio terms: several studies report a steeper age-related climb in MetS odds for women than men [54,55], and older women appear to carry a heavier burden of the cardiometabolic and cardiovascular consequences that follow [51,58].

Not everyone agrees on the shape of these trends, though. Some studies find similarly steep age-related increases for both sexes; others report that the sex gap narrows, or even flips, in the oldest cohorts [51,58]. Which components drive the risk can shift too: triglyceride disorders tend to ease with age in men while holding steady or worsening in women [54,58]. And there is a deeper methodological question underneath all of this: whether applying the same diagnostic cut-offs across every age and sex group even makes sense, given how differently individual components seem to behave across demographic subgroups [17,59].

Age, sex, and gender clearly matter for MetS risk, but the biological, behavioral, and social forces behind that relationship pull in different directions depending on the population studied. That argues for age- and sex-specific approaches to diagnosis and prevention rather than one-size-fits-all criteria, and for more work aimed at resolving the inconsistencies that keep showing up across studies [51,52,54,56,58].

Race/ethnicity, education, marital status, household size, and the poverty–income ratio each leave a mark on MetS risk, though not always in a straightforward or consistent way. Non-Hispanic Black, Hispanic, and some Asian subgroups tend to show higher MetS prevalence and severity than non-Hispanic Whites even once socioeconomic and lifestyle factors are accounted for [52], but the pattern is not uniform within groups either. Non-Hispanic Black men, for instance, sometimes show lower MetS prevalence than White men, while Black women show higher rates than White women [52]. These disparities persist across the life course, and environmental or behavioral factors alone do not fully explain them [52,60].

Socioeconomic status generally tracks with MetS risk the way you would expect: lower education and income mean higher risk, particularly for women and certain ethnic groups [60]. But the relationship bends depending on who you are looking at. Higher income protects non-Hispanic Whites and Asians from MetS, yet among non-Hispanic Blacks it is sometimes neutral, or even linked to higher obesity rates [52,60]. Curiously, how people perceive their own social standing predicts MetS severity better than their actual income or education does, especially for women [57,61].

Household-level social variables are harder to pin down than income or education. The evidence base for factors such as marital status and household composition is thin and inconsistent, and this proposal therefore carries them as descriptive covariates rather than as hypothesised risk factors. What does replicate is that where people place themselves on a social ladder tracks metabolic risk over and above their measured income and education, and does so most strongly in women [57,61].

Lower education, lower income, and certain racial or ethnic backgrounds line up fairly reliably with higher MetS risk, but subgroup, sex, and context all bend the size and even the direction of that relationship. Prevention strategies built around a single population-wide template are likely to miss a lot of this variation [52,57,60].

BMI and hip circumference remain the default anthropometric tools in MetS research. Both track reliably with MetS as they rise, along with individual components like hypertension, dyslipidemia, and hyperglycemia [17,62]. BMI alone keeps underperforming central obesity measures such as waist circumference or waist-to-hip ratio, though, largely because it cannot tell where the fat actually sits [62].

Fold hip circumference into a ratio or composite index, though, and its predictive power improves: waist-to-hip ratio and the body adiposity index both outperform plain BMI at identifying MetS risk, especially in older adults and women [62,63]. On its own, hip circumference has occasionally shown an inverse relationship with metabolic risk, possibly because gluteofemoral fat is somewhat protective, but that finding does not hold everywhere [64,65].

Which anthropometric measure predicts MetS best is still an open question. Some studies find BMI and hip circumference (or their derived indices) roughly equivalent; others find central obesity measures clearly superior, particularly in men and certain ethnic groups [62-64]. Cut-off values complicate things further, since the optimal thresholds for BMI and hip circumference shift by age, sex, and ethnicity [59,62]. The point applies across populations and not only across individuals: thresholds validated in European-ancestry samples misclassify cardiometabolic risk when applied unchanged to Asian populations, where risk begins to rise at a lower body mass index [59].

BMI and hip circumference both associate with MetS, but how well either one predicts it depends heavily on context. Central obesity measures, and composite indices that combine waist and hip data, generally discriminate MetS risk better. Whether hip circumference itself reads as protective or risk-enhancing seems to come down to how it is measured and in whom, a reminder that anthropometric screening probably needs to be tailored rather than standardized [59,64,65].

Smoking, alcohol, and physical activity relate to MetS in ways that do not fit a single pattern: some links hold up consistently across populations, others do not.

Physical activity is the clearest signal of the three. Low or absent activity shows up repeatedly across large studies and meta-analyses as an independent risk factor for MetS, while people who are consistently or newly active carry lower risk [23,24,66]. That protective effect holds across sexes and age groups, and more active people tend to show better numbers across the board: lower blood pressure, triglycerides, and waist circumference [23,24].

Smoking is murkier. Control for other lifestyle factors and the smoking–MetS association disappears in some studies, particularly among women [67]. Other research, especially in male samples, finds the opposite: smoking raises MetS risk along with specific components like low HDL, high triglycerides, and abdominal obesity [67,68]. Pair smoking with low physical activity and the risk climbs further. How much depends on intensity and which subgroup you are looking at [67,68].

Alcohol is the messiest variable of all. Plenty of studies find no significant alcohol–MetS link [69]. Others report that heavy or frequent drinking raises risk, while light-to-moderate drinking looks neutral or even protective [69]. Differences in drinking pattern, in the sex composition of the samples, and in how the reference group is defined plausibly account for much of that disagreement, which makes population-level summaries of alcohol and MetS difficult to interpret [67,69].

Physical activity is the one lever here that behaves predictably and is genuinely modifiable; smoking and alcohol swing around depending on sex, intensity, and lifestyle context. Any prevention strategy probably needs to lean hard on activity promotion while treating smoking- and alcohol-related interventions as population-specific problems rather than universal ones [23,68,69].

Heavy metals complicate the picture further. Lead, cadmium, mercury, selenium, and manganese have all been studied against MetS, and again the findings split between consistent and contradictory.

Cadmium and mercury show the most consistent pattern: most meta-analyses and large observational studies link higher blood levels of both to greater MetS risk and to specific components like abdominal obesity, hypertension, and dyslipidemia [7-11,70]. Selenium is an odd case: it is an essential nutrient, yet elevated blood levels, especially alongside zinc, still track with higher MetS prevalence [12,13,35]. Lead shows up too, with some studies linking it to increased MetS risk, particularly in men, and to higher blood pressure, triglycerides, and waist circumference [10,27,29,36].

Several large cross-sectional studies and meta-analyses report inverse or null associations for lead, cadmium, and mercury in some populations, so the direction of association is not settled [8,16,38]. United States analyses have reported associations differing from those in Asian populations in direction as well as magnitude, rather than weaker versions of the same effect [33,34]. Manganese is the least settled of the group: increased risk shows up only in specific age brackets in some studies, and not at all in others [15,16].

The mechanistic story that has emerged points to oxidative stress, inflammation, and disrupted lipid and glucose metabolism as the pathways connecting these metals to MetS [13,32,71]. But co-exposure to multiple metals, genetic background, and lifestyle factors like diet and smoking all modify the relationship, which is probably why results look so population-specific [33-35]. Sex differences complicate things further: some metals associate more strongly with MetS in men than in women [14,70].

Cadmium and mercury carry the most consistent evidence for increased MetS risk, selenium less so; lead and manganese remain mixed and heavily context-dependent. Population, exposure level, and confounding all seem to be doing real work behind that variability, which is exactly the gap prospective and mechanistic studies still need to close [7,8,72].

3. Method

3.1. Machine Learning

3.1.1. Data sources

Six consecutive NHANES cycles feed this study: 2007–2008, 2009–2010, 2011–2012, 2013–2014, 2015–2016, and 2017–2018. Pooled together, they use a stratified, multistage probability design built to produce nationally representative estimates for the U.S. civilian, non-institutionalized population. Merging the examination, laboratory, questionnaire and demographic files across all six cycles gives a pooled sample of 59,842 individuals, of whom 34,770 are adults aged 20 and older. Pregnant participants (372) are excluded, because pregnancy changes waist circumference, blood pressure, lipids and glucose. Anyone missing one of the five MetS components (waist circumference, fasting glucose, HDL cholesterol, triglycerides, or blood pressure) is excluded next (20,653), mostly because triglycerides and glucose are measured only in the morning fasting subsample; then participants who did not meet the fasting criterion of 8 to 24 hours (695, identified by a zero fasting-subsample weight); and finally anyone with no blood-metal measurement at all (2,171). A participant with, say, lead and cadmium measured but not the other three is retained. The final analytic cohort is 10,879 adults (Figure 1, Table 2).

Because the analytic cohort is, by construction, the fasting subsample, survey-weighted descriptive statistics use the fasting-subsample weight (WTSAF2YR) divided by the number of pooled cycles, following the NHANES analytic guidelines. In 2013–2014 and 2015–2016 the metals panel was measured in a separate one-half subsample, and NCHS provides no weight for the overlap of the two subsamples, so metal estimates for those cycles are approximate; this is stated as a limitation. Standard errors account for the stratified, multistage design via Taylor-series linearization (the ultimate-cluster method) using the primary sampling unit (SDMVPSU) and stratum (SDMVSTRA) variables, as recommended for the masked public-use design. Models are fitted unweighted, since the target is individual-level discrimination rather than a population parameter.

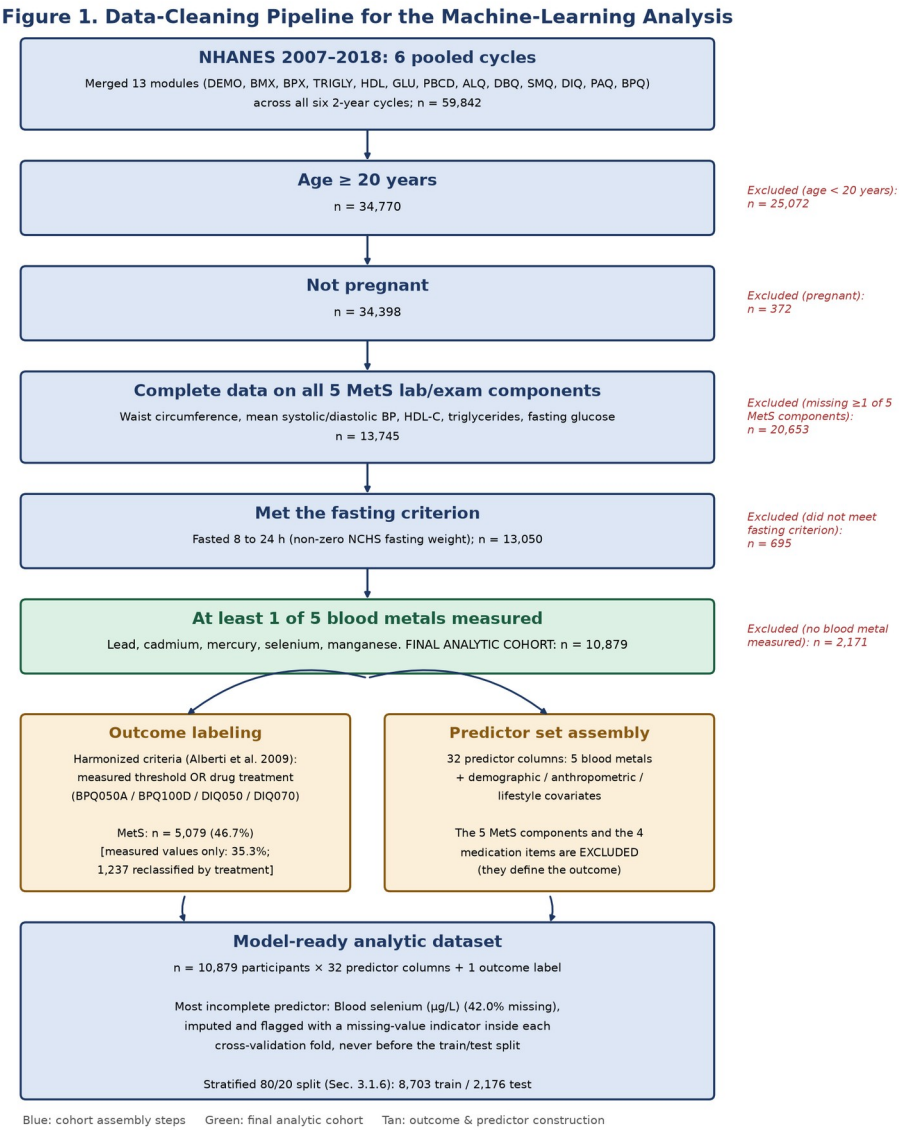


Figure 1. Data-cleaning pipeline used for the machine-learning analysis.

3.1.2. Diagnostic criteria

Metabolic syndrome is defined by the harmonized criteria of the 2009 joint interim statement (Alberti et al., Circulation 2009;120:1640–1645), which retain the 2005 AHA/NHLBI thresholds and drug-treatment clauses (Grundy et al., Circulation 2005;112:2735–2752). Each component is a yes/no flag scored positive if the measured value crosses the threshold, abdominal obesity (waist circumference ≥ 102 cm in men, ≥ 88 cm in women), hypertriglyceridemia (triglycerides ≥ 150 mg/dL), low HDL-C (< 40 mg/dL in men, < 50 mg/dL in women), elevated blood pressure (mean systolic ≥ 130 mmHg or mean diastolic ≥ 85 mmHg, averaged over all valid readings), or impaired fasting glucose (≥ 100 mg/dL), OR if the participant reports current drug treatment for that condition: antihypertensive medication (BPQ050A) for blood pressure, insulin or oral diabetes medication (DIQ050, DIQ070) for glucose, and cholesterol-lowering medication (BPQ100D) for the two lipid components. NHANES does not name the cholesterol drug, whereas the guideline's lipid clause refers to drugs such as fibrates and nicotinic acid, so applying

BPQ100D to both lipid components is a stated modelling choice that is tested in a sensitivity analysis. Three or more positive components define MetS. The four medication items and the five component measurements build the label only and are excluded from the predictor set (section 3.1.3). The treatment clauses reclassify 1,237 of 10,879 participants from MetS-negative to MetS-positive (none move the other way), raising prevalence from 35.32% (measured values only) to 46.69% (harmonized definition). Sensitivity analyses (section 3.1.7) re-fit the leading models with the measured-value-only label, without the lipid-medication clause, and with blood pressure from the first reading only.

3.1.3. Variable definition and predictor selection

The predictor set is built to avoid a specific problem: several of the candidate risk factors discussed elsewhere in this proposal are also among the five variables used to define MetS status itself. Waist circumference, blood pressure, HDL cholesterol, triglycerides, and fasting glucose are pulled from NHANES for one purpose only: determining who is and is not classified as having MetS. None of them re-enter the model as predictors. Including them as inputs to a model that predicts a label built from those same five variables would let the model partially reconstruct its own answer key rather than learn a genuine risk relationship. Avoiding that circularity is a deliberate design choice throughout this pipeline, not an afterthought.

The predictor set draws on four categories. Blood metals anchor the exposure side: lead (LBXBPB), cadmium (LBXBCD), total mercury (LBXTHG), selenium (LBXBSE), and manganese (LBXBMN), all measured in the same whole-blood laboratory module. Demographic and socioeconomic covariates cover age (RIDAGEYR), sex (RIAGENDR), race/ethnicity (RIDRETH1), education (DMDEDUC2), marital status (DMDMARTL), household size (DMDHHSIZ), the income-to-poverty ratio (INDFMPIR), and examination period (RIDEXMON). Body mass index (BMXBMI) is the sole anthropometric predictor; hip circumference was dropped after a collinearity screen on the training partition only found it correlated with BMI at r = 0.932 and missing in 81.3% of the training partition. Because BMI is strongly correlated with waist circumference, which defines one MetS component, part of BMI's predictive weight is close to definitional; a model without BMI is fitted for comparison. Lifestyle covariates are smoking history and current intensity (SMQ020, SMQ040; never-smokers are coded as not smoking now, since NHANES skips them past SMQ040), alcohol (any alcohol in the past 12 months, built the same way in every cycle from the past-year drinking frequency, and average drinks per drinking day, ALQ130, set to zero for past-year non-drinkers), self-rated diet quality (DBQ700), vigorous work, vigorous recreational and moderate recreational activity (PAQ605, PAQ650, PAQ665), active-transportation minutes (PAD645, zero when the gating question PAQ635 is answered no), and resting pulse (BPXPLS). Refusal and don't-know codes are recoded to missing using each item's own code width, so genuine answers such as 7 or 9 drinks are kept. Categorical variables are one-hot or ordinally encoded; continuous variables are median-imputed and standardized, and each continuous predictor with missing values carries a binary indicator (1 = missing), with one shared indicator for selenium and manganese, which were not assayed in 2007–2010. All imputation, scaling and resampling happens inside the model-fitting pipeline (section 3.1.6), never beforehand, so no information from a held-out fold or the test set leaks into training.

3.1.4. Baseline analysis

The baseline analysis follows the NHANES complex-survey protocol. Using the fasting-subsample weight divided by the number of cycles (section 3.1.1), survey-weighted domain means are computed separately for the MetS and non-MetS groups for age, BMI and the income-to-poverty ratio, and weighted geometric means for the five blood metals, which are strongly right-skewed. Weighted percentages are computed for sex, race/ethnicity, education and smoking history. Standard errors use Taylor-series linearization with domain estimation; the difference between groups is linearized directly, so the covariance between the two domains is included, and is tested with a design-based t test (on the log scale for the metals) or, for categorical variables, the design-based adjusted Wald F test, with degrees of freedom equal to the number of primary sampling units minus the number of strata.

3.1.5. Feature selection

Feature selection here is knowledge-driven rather than algorithmic: there is no LASSO or stepwise procedure deciding which variables enter the model. Every candidate predictor described in section 3.1.3 is chosen ahead of time on clinical and epidemiological grounds, then carried through imputation, standardization, and categorical encoding as a fixed set. What SHAP and permutation importance contribute afterward is a ranking, not a selection: they quantify how much each already-chosen predictor actually moves the trained model's output, and how consistently, but they never add or remove a variable from consideration. Two importance measures are computed independently and then compared against each other (SHAP values from the best-performing tree-based model by test AUC, and model-agnostic permutation importance across the top several models), so that a feature judged important because of a genuine, robust signal in the data can be distinguished from one that only looks important to a single algorithm's particular quirks.

3.1.6. Modeling

The cleaned NHANES cohort is split 80/20 into training and hold-out test sets, stratified on MetS status, to check out-of-sample performance; the test set is set aside immediately after cleaning and is not touched again until final evaluation. Eleven algorithms are trained: logistic regression, decision tree, random forest, gradient boosting, LightGBM, XGBoost, CatBoost, support vector machine, Gaussian Naive Bayes, multilayer perceptron, and k-nearest neighbors, each predicting the binary, medication-adjusted MetS label from the panel of metal, demographic, anthropometric, and lifestyle predictors described above, never from the biomarkers or medication-use variables that define MetS itself. Hyperparameters for each algorithm are tuned over a small grid (at most 12 configurations, searched exhaustively) under stratified 10-fold cross-validation on the training partition only, scored on ROC AUC. Class imbalance (46.69% MetS prevalence under the harmonized outcome definition) is handled with SMOTENC rather than plain SMOTE, because the feature matrix mixes continuous measurements with one-hot and binary categorical variables; SMOTENC generates synthetic continuous values by interpolation as SMOTE would, but assigns categorical values by majority vote among nearest neighbors instead of interpolating them, which avoids manufacturing fractional, non-existent category values (e.g., a synthetic "sex" of 0.4) that plain SMOTE would otherwise produce on a mixed feature space. SMOTENC is refit fresh inside every cross-validation fold and inside the final training fit, never applied once to the whole dataset beforehand, so that synthetic minority-class samples are always generated from training data alone. Each continuous predictor with missing values also carries a missing-value

indicator (see section 3.1.3) alongside its median-imputed value. The decision threshold used for each model's sensitivity/specificity trade-off is chosen from out-of-fold predictions on the training partition via the Youden index, not from the test set, keeping every step that could leak test-set information out of the process that produces the final, once-touched evaluation. Running eleven algorithms (from simple and fully interpretable through complex ensembles) and testing all of them on the same untouched 20% of subjects means the resulting AUC, calibration, and error metrics reflect genuine generalization rather than an artifact of any one algorithm's fit to this particular sample.

3.1.7. Model evaluation and interpretation

Evaluation runs entirely in Python (3.11–3.13), using scikit-learn, imbalanced-learn, LightGBM, XGBoost, CatBoost, and SHAP, chosen for reproducibility and because the leakage-safe pipeline described above depends on scikit-learn's Pipeline object refitting preprocessing and SMOTENC inside every fold, a guarantee that is harder to enforce across a mixed-language workflow. Feeding the 20% test set into each fitted pipeline generates ROC curves and confusion matrices for every classifier; from those, AUC, AUPRC, Brier score, Youden's index, accuracy, sensitivity, specificity, F1 score, Cohen's kappa, positive predictive value, and negative predictive value are all derived, each with a 1,000-resample bootstrap 95% confidence interval, at both the Youden-optimal and a fixed 0.5 cutoff for transparency. Because the top few models by point-estimate AUC on a roughly 2,200-participant test set routinely sit within a fraction of a percentage point of each other (well inside plausible sampling noise), DeLong's test for correlated AUROCs compares the top-ranked model against every other model, with a paired-bootstrap confidence interval for each difference, and flags which models are not statistically distinguishable from it at the 95% level; that flag is carried through every subsequent table and figure that ranks models, so the results never present a coin-flip-sized margin as a definitive "best" algorithm. Calibration curves confirm whether predicted risk lines up with observed MetS prevalence. Interpretability comes last: SHAP supplies summary bar plots, beeswarm plots, dependence plots (including a dedicated panel comparing all five metals on a common scale, regardless of each metal's individual rank), force plots, and decision plots to quantify each feature's impact; permutation importance, computed independently across the top several models, cross-checks that ranking for agreement rather than relying on a single model's SHAP output alone.

Further checks round out the evaluation. A decision-curve analysis compares net benefit against treat-all and treat-none strategies across threshold probabilities from 2% to 80%. A subgroup analysis recomputes discrimination and calibration within sex, race/ethnicity and age-tertile strata. Sensitivity analyses on the three leading models, each with its tuned hyperparameters, add a diagnosed-diabetes indicator, replace SMOTENC with class weighting, restrict the metal panel to lead, and substitute three alternative outcome definitions (measured values only, no lipid-medication clause, first blood-pressure reading only). Research question Q3 is answered directly by refitting the leading model without the five metals, with lead as the only metal, and without BMI, and by a logistic regression on age and BMI alone, each compared with the full model on the same test participants by DeLong's test. Repeated stratified 10-fold cross-validation (five repeats) shows how much performance varies between data partitions, and a temporal validation re-tunes and refits the leading algorithm on the 2007–2014 cycles and tests it on 2015–2018.

3.2. Mendelian Randomization

3.2.1. Study design

This MR study tests whether blood metal exposures, together with any non-metal factor the machine-learning phase identifies as a comparably strong predictor, have a causal effect on the individual components of metabolic syndrome in individuals of predominantly European ancestry (Figure 2), using publicly available summary-level GWAS data; Table 3 summarizes the instrument–exposure–outcome structure.

Exposures are chosen by two criteria applied jointly. The first is the machine-learning evidence (section 3.1.7). In the final run on the 10,879-participant cohort, body mass index and age dominated the ranking; the five metals sat in the middle and lower part of it, and removing all five changed test-set discrimination by −0.0001 AUROC units (DeLong p = 0.951). The metal evidence, from strongest to weakest by mean rank across the four leading models, was: lead (permutation rank 11 of 32 in CatBoost, above its permutation standard deviation; 6 to 11 across the four leading models); selenium (permutation rank 16 of 32 in CatBoost, within its permutation standard deviation; 9 to 16 across the four leading models); cadmium (permutation rank 13 of 32 in CatBoost, above its permutation standard deviation; 13 to 16 across the four leading models); mercury (permutation rank 15 of 32 in CatBoost, within its permutation standard deviation; 15 to 30 across the four leading models); manganese (permutation rank 17 of 32 in CatBoost, within its permutation standard deviation; 17 to 32 across the four leading models). Lead, the most stable metal across models, and selenium, the next most stable, are the primary metal exposures. Cadmium is secondary: its signal was distinguishable from noise in CatBoost but less stable across models, and it has the strongest prior literature and a published never-smoker instrument. The second criterion is a usable genetic instrument under the rule in section 3.2.3, applied identically to every exposure. An associational ranking and a causal test answer different questions: a factor can be predictive without being causal, or causal without being the strongest predictor in observational data, so exposures with a weaker ML signal but a usable instrument are retained as exploratory. Table 1 lists the candidate exposures against the evidence for each.

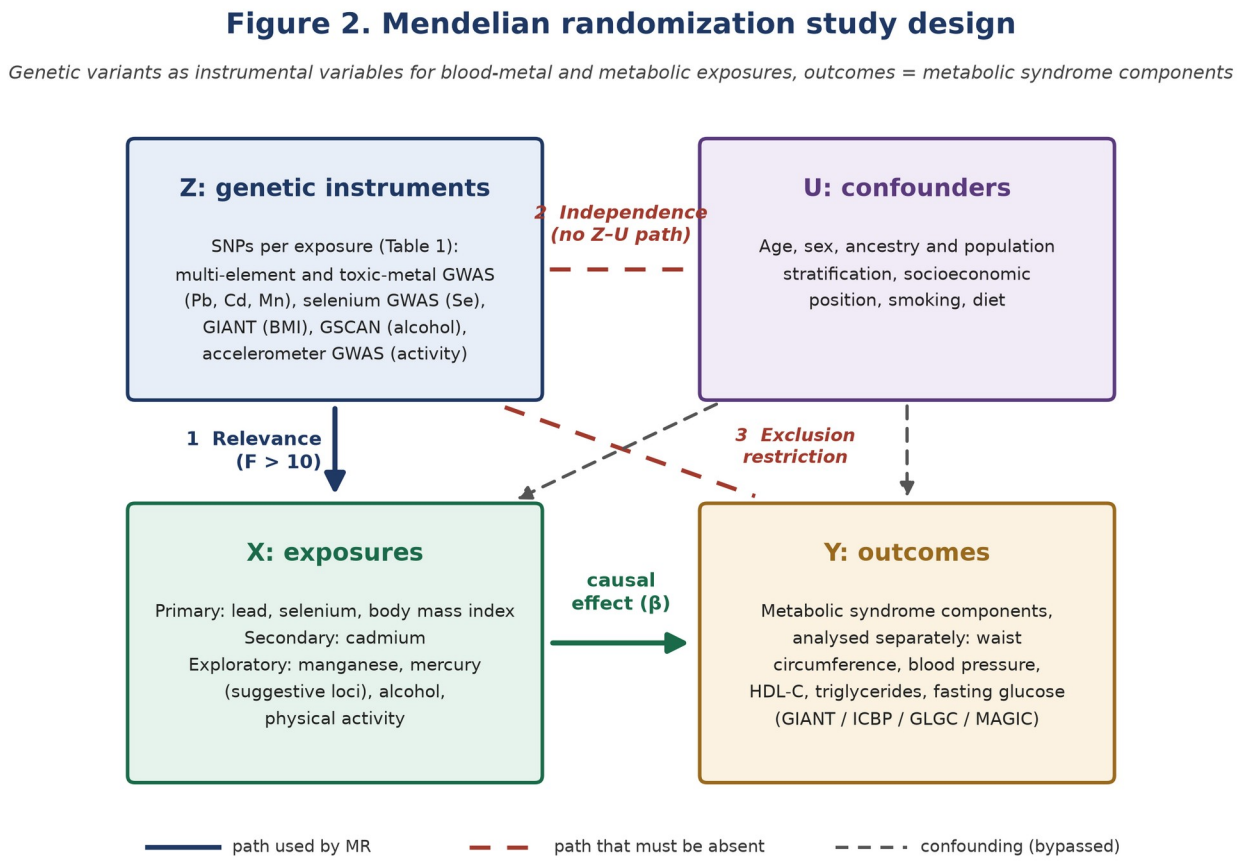


Figure 2. Study design: genetic variants serve as instrumental variables linking each blood-metal or metabolic exposure to the metabolic-syndrome outcome, subject to the three core IV assumptions (relevance, independence, exclusion restriction).

3.2.2. Data sources

GWAS summary statistics are drawn from the IEU OpenGWAS project (https://gwas.mrcieu.ac.uk/) and, where a trait is not indexed there, directly from the original publication's supplementary data. Independent datasets drawn from populations of matched ancestry are selected wherever possible to limit confounding in the genetic associations, and a two-sample MR design is implemented separately for each exposure–outcome pair. Where the exposure and outcome GWAS share participants (for example UK Biobank contributes to both the BMI and some blood-pressure GWAS), sample overlap is reported and a non-overlapping outcome source is preferred.

For the metals, exposure instruments come from genome-wide meta-analyses of blood metal concentrations: lead, cadmium and manganese from Ng et al.'s toxic-metals GWAS and the multi-element GWAS by Moksnes et al. [73,74], cadmium also from the never-smoker cadmium GWAS by Borné et al. [75], and selenium from Cornelis et al.'s GWAS of blood selenium concentrations [76]. For mercury, the largest GWAS of blood mercury to date, in pregnant women and children of European ancestry, reported suggestive but not genome-wide-significant loci [77]; mercury is therefore analysed

as an exploratory exposure under the relaxed threshold of section 3.2.3, and only if its instrument passes the strength check. Body mass index draws on the GIANT consortium meta-analyses [78,79], alcohol consumption on the GSCAN consortium [80], and physical activity on the UK Biobank accelerometer-derived GWAS [81]; the latter measures overall device-measured activity, whereas the machine-learning signal came from self-reported vigorous recreational activity, so the two constructs are not identical.

On the outcome side, because a dedicated, adequately powered GWAS of the composite MetS phenotype is not consistently available across ancestry-matched cohorts, the primary outcome data are drawn from GWAS of the individual MetS components instead: lipids from the Global Lipids Genetics Consortium, fasting glucose from the MAGIC consortium, blood pressure from the International Consortium of Blood Pressure GWAS or UK Biobank, and waist circumference from GIANT, run as parallel analyses rather than pooled into a single composite outcome. This avoids diluting a heterogeneous phenotype and keeps the results interpretable against the same clinical criteria used to define MetS in the machine-learning phase.

3.2.3. Genetic instrument selection

Instrument strength is checked for every exposure with the F-statistic ($F = \beta^2/SE^2 > 10$), because weak instruments bias two-sample estimates towards the null [82,83]. One rule applies to every exposure: SNPs are selected at genome-wide significance ($P < 5 \times 10^{-8}$), and when fewer than three independent SNPs reach it, the threshold is relaxed to $P < 5 \times 10^{-6}$ and the exposure is labelled as using suggestive instruments. Under this rule mercury is analysed rather than excluded, as exploratory; its result is reported separately from the primary analysis. LD clumping ($r^2 = 0.001$ within a 10,000 kb window) using an ancestry-matched reference panel retains the lowest-p-value SNP at each locus [49,84], and SNPs associated with likely confounders or with the outcome are flagged and removed using PhenoScanner [85] or, if that service is unavailable, the NHGRI-EBI GWAS Catalog. During harmonization (action = 2), palindromic SNPs with intermediate allele frequencies (0.40–0.60) are excluded, along with any SNP whose minor allele frequency falls below 0.01 [84,86].

3.2.4. Statistical analysis

The methods used for each exposure–outcome pair depend on how many SNPs the instrument has, because most robust MR estimators need several. With one SNP, the Wald ratio is the only estimate. With two or more, the inverse variance weighted (IVW) estimator is primary, combining SNP-specific Wald ratios under the assumption that no pleiotropy is present [87]; when heterogeneity and pleiotropy are absent, maximum likelihood produces equivalent point estimates with tighter standard errors [87]. With three or more SNPs, MR-Egger regression, under the Instrument Strength Independent of Direct Effect (InSIDE) assumption, detects directional pleiotropy through its intercept [88], and the weighted median remains valid if up to half the instruments are invalid [89]; the weighted mode adds robustness against certain InSIDE violations [90].

Heterogeneity in the IVW estimates is assessed with Cochran's Q (two or more SNPs) and, for MR-Egger, with Rücker's Q [91]; the I^2GX statistic checks the no-measurement-error assumption on which MR-Egger relies [92], and the MR-Egger intercept provides a further check for pleiotropy [88].

With ten or more SNPs, sensitivity analyses back this up: MR-PRESSO flags and removes outlier SNPs [93]; RadialMR, Cook's distance and Studentized residuals identify variants with outsized influence [94]; and a leave-one-out analysis checks whether the result hinges on a single SNP [84]. For exposures with fewer SNPs these methods are undefined or unstable and are not reported.

Where enough SNPs are available, alternative estimators robust to pleiotropy are run for comparison: MR-Lasso [95], constrained maximum likelihood [96], contamination mixture models [97] and the mode-based estimator [90]. The Steiger directionality test confirms which way the causal arrow points for each exposure–outcome pair, which matters in particular for body mass index and the metabolic syndrome components, where reverse causation is biologically plausible [98]. The analyses will be reported in accordance with the STROBE-MR guideline [99].

All of this runs in R version 4.3.1, using packages including TwoSampleMR, MendelianRandomization, MR-PRESSO, RadialMR and MRMix, with additional analyses in Stata version 17 via the mrrobust package. Because every outcome is a continuous component trait, results are reported as beta coefficients (change in the outcome per standard-deviation or unit change in the genetically predicted exposure) with 95% confidence intervals, and associations with $p < 0.05$ are treated as statistically significant, with the number of exposure–outcome pairs reported alongside so that multiplicity can be judged (Figure 2).

Table 1. Candidate MR exposures identified by the machine-learning phase, screened against published genetic instrument availability.

Exposure	AI-phase signal	Genetic instrument	MR status
Body mass index	SHAP rank 1 and permutation rank 1 of 32; removing it changed test AUROC by −0.0796. Part of its weight is definitional, because waist circumference defines one MetS component	GIANT consortium meta-analyses, ~700,000 participants [78,79]	Primary exposure
Lead	Most consistent metal: permutation rank 11 of 32 (above its permutation SD), ranks 6 to 11 across the four leading models; SHAP rank 18; SHAP contribution falls with concentration	Loci for blood/erythrocyte lead in the toxic-metal and multi-element GWAS [73,74]; strength to be confirmed (F > 10)	Primary exposure
Selenium	Permutation rank 16 (not above its permutation SD), ranks 9 to 16 across models; SHAP rank 16; SHAP contribution rises with concentration	GWAS of blood selenium concentrations, loci at 5q14 and 21q22.3 [76]	Primary exposure
Cadmium	Permutation rank 13 (above its permutation SD), SHAP rank 15; U-shaped SHAP contribution; retained for its strong prior literature	Toxic-metal and multi-element GWAS [73,74]; never-smoker cadmium GWAS [75]	Secondary exposure
Manganese	Permutation rank 17, SHAP	Two genome-wide-significant loci (4q24,	Exploratory

	rank 23; no consistent SHAP direction	1q41) [73,74]	
Alcohol consumption	Drinks per drinking day: SHAP rank 8, permutation rank 6	GSCAN consortium, ~1.2 million participants [80]	Exploratory
Physical activity	Vigorous recreational activity: SHAP rank 3, permutation rank 4; the GWAS measures device-measured overall activity, a different construct	UK Biobank accelerometer-derived GWAS, 14 loci [81]	Exploratory
Mercury	Permutation rank 15, SHAP rank 19; SHAP contribution falls with concentration	Suggestive (not genome-wide-significant) loci only [77]; relaxed threshold (section 3.2.3)	Exploratory (suggestive instrument)

Table 2. Participant flow for the machine-learning analytic cohort.

Step	N remaining	Excluded at this step
0. NHANES 2007–2018 pooled sample	59,842	–
1. Age ≥ 20 years	34,770	25,072 (age < 20)
2. Not pregnant	34,398	372 (pregnant)
3. Complete data on all 5 MetS components (waist, blood pressure, HDL, triglycerides, glucose)	13,745	20,653 (missing ≥ 1 MetS component)
4. Met the fasting criterion (8 to 24 h)	13,050	695 (not fasted)
5. At least 1 of 5 blood metals measured	10,879	2,171 (no metals measured)
6. Final analytic cohort (MetS status computed)	10,879	0

Table 3. MR instrumental-variable design (summary).

Instrumental variable	Exposure(s)	Outcome(s)
Genetic variants (metal- and trait-specific instruments, Table 1)	Blood lead, selenium and body mass index (primary); cadmium (secondary); manganese, mercury (suggestive instruments), alcohol consumption, physical activity (exploratory)	Metabolic syndrome components, analysed separately: waist circumference, triglycerides, HDL-C, blood pressure, fasting glucose (GIANT / GLGC / MAGIC / ICBP, as applicable)

4. Hypotheses and Research Questions

4.1. Machine Learning

4.1.1. Hypotheses

H1: Machine learning models trained on NHANES data will predict MetS presence in adults more accurately than traditional regression-based approaches.

H2: Ensemble-based models such as LightGBM, XGBoost, CatBoost, and Random Forest will outperform simpler models like logistic regression or decision trees.

H3: The five blood metals, considered jointly, will add discrimination beyond demographic, anthropometric and lifestyle variables (tested by research question Q3).

H4: The relative importance of the individual metals will differ between importance methods and between algorithms, so only metals whose importance is consistent across both will be prioritized for the MR phase.

4.1.2. Research questions

Q1: How reliably can machine learning models trained on NHANES data (2007–2018) predict MetS?

Q2: Among the candidate algorithms (logistic regression, LightGBM, XGBoost, CatBoost, Random Forest, Gradient Boosting, SVM, MLP, and others), which provides the best balance of predictive accuracy, sensitivity, specificity, and calibration?

Q3: How much does adding the five-metal exposure panel improve predictive accuracy over models built solely on demographic, anthropometric, and lifestyle variables?

Q4: What do SHAP- and permutation-importance-based feature rankings reveal about each metal's contribution relative to conventional MetS risk factors, and how consistent is that ranking across independently trained models?

4.2. Mendelian Randomization

4.2.1. Hypotheses

H5: Publicly available GWAS will yield valid instruments (F > 10) for blood lead, selenium and body mass index; instruments for cadmium, manganese, alcohol consumption and physical activity may be fewer or weaker, and mercury has suggestive loci only.

H6: The instruments for the primary exposures (lead, selenium, body mass index) will show sufficient strength (F > 10) for reliable causal inference, while instrument strength for the secondary and exploratory exposures is expected to vary.

H7: Where an exposure has enough SNPs, heterogeneity and pleiotropy will be detectable and robust methods (weighted median and MR-Egger with three or more SNPs, MR-PRESSO with ten or more) will be used to limit the resulting bias; with fewer SNPs only Wald-ratio or IVW estimates will be interpreted, cautiously.

H8: Genetically predicted blood selenium and lead will be associated with one or more MetS components (blood pressure, fasting glucose, triglycerides, HDL-C). The effect of lead is tested two-sided, because the machine-learning phase showed an inverse cross-sectional association that may reflect confounding; cadmium is tested as a secondary exposure.

H9: Exposures ranked highest by the machine-learning phase (body mass index in particular) will show the strongest and most consistent causal estimates, while any exposure with a weak machine-learning signal but an adequate instrument will help distinguish a genuinely non-causal association from one the modeling phase under-detected.

4.2.2. Research questions

Q5: How many SNPs qualify as valid instruments for each candidate exposure (lead, selenium, body mass index, cadmium, manganese, mercury, alcohol consumption, physical activity), and at which threshold?

Q6: Do these SNPs satisfy the three core MR assumptions (relevance, independence, and exclusion restriction) for each exposure?

Q7: How much heterogeneity exists among the SNP-specific causal estimates for each exposure–outcome pair, as assessed via Cochran's Q and I^2?

Q8: For exposures with enough SNPs, do the MR-Egger intercept and MR-PRESSO tests show evidence of directional pleiotropy in any exposure–outcome association?

Q9: Does the MR analysis support a causal effect of any blood-metal exposure on the MetS components, and if so, does the ranking of causal effect sizes across metals resemble the ranking produced by the machine-learning phase?

Q10: For exposures analysed with suggestive instruments (mercury) or as exploratory (manganese, alcohol consumption, physical activity), does subsequent evidence (an updated GWAS or the sensitivity analyses in section 3.2.4) change their classification?

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